

Timothy Tarter

Undergraduate at Department of Mathematics, James Madison University
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Research Interests

My primary research interest are algebraic geometry & combinatorial geometry. I am currently working on a project in Schubert Calculus & Equivariant Cohomology with Dr. Rebecca Field. Previously, I have conducted minor independent research projects in measure theory & dynamical systems.

Education

B.S. in Mathematics, Minor in Computational Analytics Fall 2023–Fall 2026

James Madison University

Deans List:

- Fall of 2023, 2024, & 2025
- Spring 2024
- Summer 2024.

Conference Talks & Preprints

- Santos & Tarter. *Modeling the Impact of NATO Policy Actions on Online Threat Populations via the Lotka-Volterra Framework*. Graduate Society for International Studies (2025). Preprint, arXiv: <https://arxiv.org/abs/2508.01904>
- Tarter. *Applications of Benford's Law Generalized to Geospatial Threats and Social Network Investigations (2025)*. Preprint, arXiv: <https://arxiv.org/abs/2501.10460>

Talks & Presentations

- *A Faster Attack on LFSR Recurrences: an Alternative to the Berlekamp-Massey Algorithm*, Joint Mathematics Meetings (2026).
- *Zariski Continuity as a Bridge Between Topology and Geometry*, James Madison University (2025).
- *A Faster Attack on LFSR Recurrences: an Alternative to the Berlekamp-Massey Algorithm*, Gannon University's Applied Intelligence Conference (2025).
- *On Y-Delta and X-Square Graph Reductions and the Study of their Algebraic Properties*, James Madison University, Department of Mathematics (2025).
- *Modeling the Impact of NATO Policy Actions on Online Threat Populations via the Lotka-Volterra Framework*, Graduate Society for International Studies (2025).

- *Applications of Benford's Law Generalized to Geospatial Threats and Social Network Investigations*, Joint Mathematics Meetings (2025).
- *Applications of Benford's Law Generalized to Geospatial Threats and Social Network Investigations*, Shenandoah Undergraduate Mathematics Symposium (2024).

Teaching Experience

Teaching Assistant, Abstract Algebra I Spring 2026

James Madison University

Led problem sessions; held office hours; wrote and administered practice exams.

Teaching Assistant, Linear Algebra Spring 2026

James Madison University

Wrote and graded quizzes and exams; held office hours.

Teaching Assistant, Real Analysis I Spring 2025

James Madison University

Led problem sessions; held office hours; wrote and administered practice exams.

Teaching Assistant, Calculus III Fall 2024

James Madison University

Led problem sessions; held office hours; wrote and administered practice exams.

Seminars

Algebraic Geometry Seminar Fall 2025–Present

Unaffiliated

Organized and led two separate seminars of two and seven students (respectively) working through Vakil's "The Rising Sea: Foundations of Algebraic Geometry." Lectures available at,

- (Primary) <https://www.youtube.com/@TimothyTarter>
- (Secondary) <https://www.youtube.com/@BaseChangeRequired>.

Algebraic Topology Seminar Spring 2026

Unaffiliated

Organized and led a seminar of three students working through Hatcher's "Algebraic Topology." Lectures available at:

- <https://www.youtube.com/@TimothyTarter>.

Abstract Algebra Seminar Fall 2026

Unaffiliated

Organized and led a seminar of ten students working through Artin's "Algebra." The lecture recordings are not publicly available at the moment.

Awards & Honors

- Brakke-Mason Research Grant, awarded \$1900 to research the Bruhat order on the set of fixed point free involutions of the symmetric group, as induced by Schubert Calculus, with Dr. Rebecca Field (2025-Present).
- Inducted to Pi Mu Epsilon Honors Fraternity (Fall 2025).

Service

- President, Madison Institute for Mathematical Finance, 2025-2026.
- President, JMU Math Club, 2024-2025.

Skills & Courses

Programming Languages

1. $\text{\LaTeX}$ - *Proficient*.
2. Python - *Proficient in Pandas, NumPy, sympy & matplotlib packages*.
3. R Studio - *Experienced in econometric modeling*.
4. Microsoft Excel - *Experienced in backtesting & financial mathematics*.
5. Maucaulay2 - *Some experience with Grobner basis computation*.
6. MATLAB - *General experience*.

Mathematics Coursework

1. Analysis I & II
2. Complex Analysis
3. Abstract Algebra I & II
4. Topology
5. Partial Differential Equations and Fourier Analysis
6. Ordinary Differential Equations
7. Linear Algebra & Advanced Linear Algebra
8. Operations Research
9. Elliptic Curve Cryptography

10. Mathematical Statistics I
11. Euclidean Geometry
12. Discrete Mathematics
13. Calculus Sequence (I-III)